

Blake Salbilla

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EDUCATION

University of Illinois Urbana-Champaign

Expected Graduation: May 2029

B.S. Computer Science, Minor in Business

GPA: 4.0/4.0

Related Coursework: Systems Programming, Database Systems, Computer Architecture, Software Design

Activities: ACM, Illinois Business Consulting, Track & Cross Country Clubs

SKILLS

Languages: Go, Python, Java, C, C++, SQL, Bash, MIPS Assembly, Verilog

Platforms/Tools: Linux, Docker, GCP, Cloudflare, Git, PostgreSQL, SQLite

PROJECTS

Self-Hosted Homelab

Present

- Designed and maintained a multi-node **Proxmox VE** hypervisor hosting OMV, Ubuntu, and containerized workloads via **Docker** and Portainer, sustaining **>99.9% uptime** across **15+** microservices.
- Architected a hybrid AI stack routing tasks between local models via **Ollama** and frontier cloud LLMs for complex reasoning, integrating an autonomous Hermes agent to execute system tasks over **iMessage**.

Cloud Cost Predictor (HackIllinois)

March 2026

- Built a full-stack financial observability platform that parsed codebases to map **20+** distinct cloud API services to real-time pricing tiers to forecast deployment expenditures.
- Aggregated multi-provider cloud resource consumption metrics into a unified dashboard and seamless API, identifying anomalous usage to prevent surprise billing spikes.

LaundryLink

January 2026

- Architected a full-stack peer-to-peer laundry platform with **Go**, **SQLite**, and **HTMX**, driving **>100** page views and user requests within the first 24 hours of launch.
- Implemented **Auth0** for user identity and **Stripe** for PCI-compliant checkout, packaging the environment with **Docker** and **Bash** deployment scripts to reduce rollout time by **50%**.

AI Socratic Tutor (Keywords AI Hackathon)

January 2026

- Engineered a full-stack tutoring system that ingested syllabi and lecture slides to ground LLM outputs, validating curriculum-accurate Socratic responses across **30+ benchmark prompts in live demos**.
- Designed high-throughput session logging with queryable search indexes, enabling educators to conduct sub-second interaction audits and detect student learning bottlenecks.

WORK EXPERIENCE

Course Assistant, CS 233

August 2026 - present

- Support **272 students** during discussions and office hours on hardware topics and digital logic.
- Guide students through debugging **Verilog**, **MIPS Assembly**, and pipelined datapaths.

Freelance Developer

July 2025 - present

- Built **4 full-stack web apps** and secure payment workflows using **Go**, **HTMX**, **Stripe**, and **GCP**.
- Automated **100+ email workflows** with **Go** and **GCP**, eliminating manual labor on status updates.